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Locating Smartphone Indoors by Using Tightly Coupling Bluetooth Ranging and Accelerometer Measurements

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Abstract: High-precision, low-cost, and wide coverage indoor positioning technology is the key to indoor and outdoor integrated location-based services, and it has broad market prospects and social value. However, achieving sub-meter level positioning accuracy in indoor environments remains a real challenge due to the blockage of indoor Global Navigation Satellite System (GNSS) signals, the complexity of indoor environments, and the unpredictability of user behavior. In this paper, we introduce a multi-module BLE broadcaster (MMBB)-based indoor positioning solution in which a tightly coupled fusion architecture is implemented on a smartphone. The solution integrates ranging measurements from multiple MMBB and the measurements of the accelerometer built into a smartphone. It becomes an instant positioning solution without any training phase by adopting a calibrated linearly segmented path loss model for ranging. We apply the pedestrian walking speed derived by the smartphone accelerometer to constrain an unscented Kalman filter method that is used to estimate the location and speed. The accuracy of the proposed method is 50% at 0.79 m and 95% at 1.6 m in terms of horizontal error distance. Position update frequency is 10 Hz and the time to first fix is 0.1 s. The system can easily adapt a global coordinator system so that it can seamlessly work together with the GNSS to form an indoor/outdoor positioning solution.

Keywords: indoor positioning; UKF; multi-module BLE; speed constraint; smartphone; accelerometer

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1. Introduction

High-precision, low-cost, and wide coverage indoor positioning technology is the key to indoor and outdoor integrated location-based services, and it has broad market prospects and social value [1]. With the ubiquity of wireless communication devices, locating a sensor-rich smartphone has become easy. However, achieving sub-meter level positioning accuracy in indoor environments remains a real challenge due to the blockage of indoor Global Navigation Satellite System (GNSS) signals, the complexity of indoor environments, and the unpredictability of user behavior.

Nowadays, smartphones integrate more and more sensors and support abundant radio frequency signals that can be used as localization sources. However, the only sensor dedicated to positioning inside a smartphone is the GNSS receiver; all other built-in sensors and RF radios, such as an accelerometer, gyroscope, magnetometer, barometer, camera, Wi-Fi, Bluetooth, and cellular wireless communication radios, are not designed for positioning [2]. For the IEEE 802.11- and 802.15.1-based indoor positioning systems, the direct physical observation is the received signal strength indicator (RSSI) [3]. This appears to be an unreliable observable for estimating positions indoors with sufficient accuracy [4]. RSSI-based positioning solutions can be classified into two approaches: trilateration using range estimation [5,6] and fingerprinting matching [7,8]. The trilateration approach uses the geometric properties of triangles to estimate a target's location [6]. This

type of approach uses RSSI to calculate the range between the transmitter and the receiver with the path loss model, and then estimates the location using the trilateration approach. Due to the fact that channel environments are severed and spatial topology is complex indoors, the signal path loss model is difficult to determine. The fingerprinting matching approach does not need the transmitter location and signal path loss model, but a complex and time-consuming offline training phase for generating a fingerprint database is needed, and the mismatching rate is high in open spaces.

Improving positioning accuracy by integrating built-in sensors (accelerometer, gyroscope, and magnetometer) has become the mainstream smartphone-based indoor positioning method [9–14]. Extended Kalman filter (EKF) [12,14], unscented Kalman filter (UKF) [15] and particle filter (PF) [16] methods are often used for integrating multiple positioning sources. The unscented Kalman filter (UKF) method is proposed as an improvement to the EKF method [17–22]. By properly weighting a finite number of points, the probability of the state distribution is propagated through the nonlinear dynamics of the system. Unscented transformation (UT) is a simple method for computing the statistics of a random variable that undergoes nonlinear transformation [23,24]. By using UT, sigma points can fully capture the true mean and covariance of a Gaussian random variable to third-order accuracy, with no Tylor expansion.

In this work, we aim to develop an MMBB-based indoor positioning system, which can achieve sub-meter level positioning accuracy at 10 Hz output frequency using a normal smartphone without any infrastructure information. The solution involves tightly coupling Bluetooth Low Energy (BLE) range measurements and the speed information derived from an accelerometer using the unscented Kalman filter method. To assess the performance of the system, we deployed a total of six MMBBs in the area of interest. Various field tests were carried out in the test area to assess the performance and verify the effectiveness of the solution.

This paper is structured as follows: A system overview of the MMBB-based indoor positioning solution is introduced in Section 2; a novel linear subsection path loss model and a range estimation method are given in Section 3.1; a pedestrian walking speed estimation method derived from a smartphone built-in accelerometer is then introduced in Section 3.2; in Section 3.3, a speed constrained unscented Kalman filter method is presented for estimating the system state, including a position vector and walking speed; the experiment is described in detail as well as numerical results are presented and discussed in Section 4; followed by conclusions drawn in Section 5.

2. System Overview

In this paper, we proposed an MMBB-based indoor smartphone positioning solution. The system tightly couples BLE ranging information derived from RSSI measurements and the speed information derived from a smartphone's built-in accelerometer using an unscented Kalman filter method. No additional hardware is needed for a normal smartphone to receive the signal and navigation messages transmitted by the MMBBs. The smartphone is capable of delivering sub-meter level positioning accuracy at 10 Hz output frequency without any knowledge of the positioning environment. As shown in Figure 1, the system architecture consists of four parts: the hardware layer, the signal transport layer, the positioning engine layer, and the application layer.

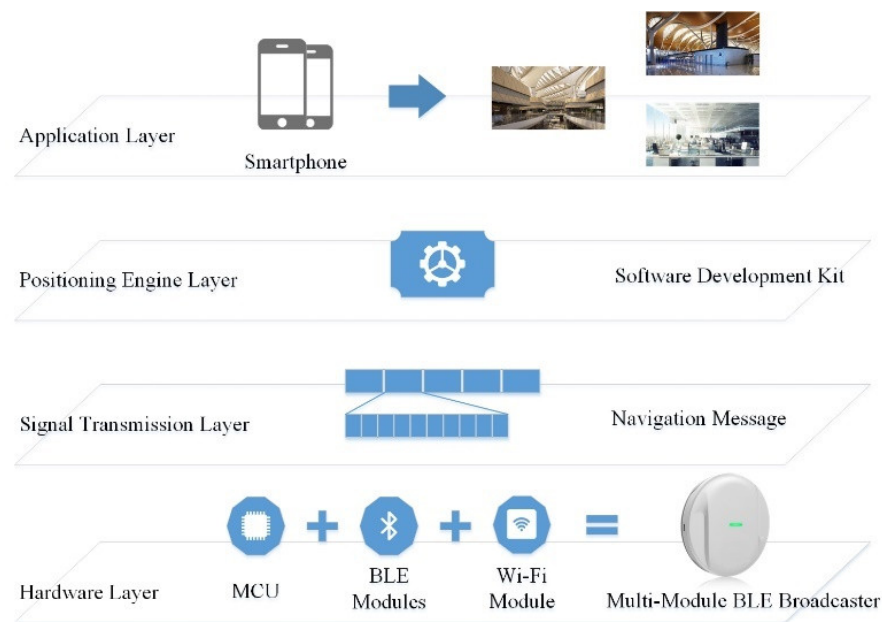


Figure 1. The architecture of the MMBB-based indoor positioning system.

- (1) Regarding the hardware layer, the MMBB consists of a micro-control unit, Bluetooth Low Energy modules, and a Wi-Fi module. The MMBB is suitable for low-power operation, which is below 100 mA in active mode and below 5 mA in stand-by mode, while providing a power supply of 2.5–5 V; the transition time from stand-by mode to active mode is neglectable.
- (2) Regarding the signal transport layer, navigation message (NM) is designed based on the Bluetooth transmission protocol. Similar to the GNSS navigation message, encrypted MMBB state and navigation information are broadcasted to users. NM provides all the necessary information to enable the user to complete the positioning task, including MMBB parameters, service parameters, and positions in the WGS84 coordinate system, which are surveyed by the total station and real-time kinematic (RTK) GPS receiver. The broadcast frequency of the NM is up to 10 Hz to support a very fast time to first fix of 0.1 s. There is no limit to the number of users theoretically, as our system is operated in a broadcast mode.
- (3) For the positioning engine layer, the core of the positioning engine is the unscented Kalman filter method that tightly couples the BLE ranging measurements and the user walking speed for estimating the user position at an output frequency of 10 Hz. As the coordinates of the MMBBs determined during the installation phase are in a global coordinate system such as GPS, the output positions of the positioning engine are also in the same global coordinate system. Specifically, for the positioning phase, we first convert the latitude, longitude, and elevation into a local ENU coordinate system to ensure no loss of accuracy and the facilitation of calculations. After the positioning is completed, the ENU coordinates are converted to latitude and longitude for output to ensure consistency with the GPS output format. Therefore, the proposed system has a significant advantage in facilitating a seamless indoor/outdoor positioning solution. The algorithms implemented in the positioning engine take advantage of these characteristics and fully support seamlessly postponing indoor/outdoor positioning. The positioning engine is finally presented as a locator server running inside the smartphone to support application development. It is like a GPS locator running in the smartphone.
- (4) The application layer is based on the positioning engine, and mobile applications can then be developed. In the space where MMBB is installed, the positioning engine works independently without the need for knowledge of the positioning environment, such as building layout, fingerprint database, and so on. From the user

perspective, it is very much like using GPS in an outdoor environment. The users only need to turn on the mobile phone, and the positioning engine will output the positions automatically to be used by multiple applications.

3. Methodology

3.1. Finite Range Estimation Based on MMBB and Segmented Linear Model

Theoretically, the energy of a radio signal attenuates along with distance from the signal transmitter. This can be expressed by using a path loss model (PLM) [25,26]. Therefore, the strength of the received signal is an indicator of its distance to the transmitter. However, the BLE signal is vulnerable and affected by multipath effects and other error sources of the signal transmission channel.

The main challenge is the sensitivity of the RSSI values which can change with environmental changes, for example, the presence of obstacles that directly lead to changes in estimating range using the signal path loss model or radio map. Intuitively, the accuracy and efficiency of range estimation, and thus positioning accuracy, is very dependent on the stability of the measured RSSI values and the path loss model. In this work, an MMBB is designed to reduce these interferences and effects and calculate accurate ranges. Based on previous research on PLM [27–29], when the geometric range between a transmitter and receiver exceeds a certain range, the accuracy of RSSI will be difficult to reflect the change in distance. Therefore, we take 10 m as the typical working distance and deployment density of MMBB, and the upper limit of range estimation for the SLM.

From a hardware perspective, the MMBB consisted of an array of BLE modules with antennas installed in a circle. The ring topological structure could ensure that the user could receive a more high-quality signal in each direction. A set of RSSI values could be measured from the antenna array and used to test each other to exclude outliers data in every positioning epoch. With this design, the stability of the ranging measurements could be improved, and the multipath effect could be reduced.

From the perspective of algorithms, we proposed a novel segmented linear path loss model (SLPLM) to calculate the range between the BLE unit and smartphone. The range is calculated as follows:

$$Dis(rssi) = \begin{cases} 0.5 & , \quad Rssi_1 < rssi \\ \frac{rssi + Rssi_2 - 2Rssi_1}{Rssi_2 - Rssi_1} & , \quad Rssi_1 < rssi \leq Rssi_2 \\ \frac{rssi + 2Rssi_3 - 3Rssi_2}{Rssi_3 - Rssi_2} & , \quad Rssi_2 < rssi \leq Rssi_3 \\ \frac{2rssi + 3Rssi_5 - 5Rssi_3}{Rssi_5 - Rssi_3} & , \quad Rssi_3 < rssi \leq Rssi_5 \\ \frac{5rssi + 5Rssi_{10} - 10Rssi_5}{Rssi_{10} - Rssi_5} & , \quad Rssi_5 < rssi \leq Rssi_{10} \\ 10 & , \quad Rssi_{10} < rssi \end{cases} \quad (1)$$

where $rssi$ represents the preprocessed RSSI. $Rssi_1, Rssi_2, Rssi_3, Rssi_5$ and $Rssi_{10}$ are calibration parameters, which are the RSSI values measured at 1 m, 2 m, 3 m, 5 m, and 10 m distances from the MMBB, respectively.

In detail, because of differences in the BLE unit hardware, every unit needed to be calibrated to obtain a more accurate range. However, even without relative movement between the transmitter and receiver, the RSSI values were not constant and could heavily oscillate. Consequently, repeated measurements and statistical methods were utilized to calibrate the parameters. The calibration field should be an open space, where the multipath effect can be lowered to a minimum. The reference points for the measured distance were 1 m, 2 m, 3 m, 5 m, and 10 m, each was measured 1000 times, after which we calculated the average μ and standard deviation σ of the RSSI values. The mean of RSSI

values within three standard deviations ($\mu \pm 3\sigma$) was used as the parameter of the corresponding distance.

Modeled on the principle of the band-pass filter, we designed a filter to estimate the range. Every 100 milliseconds, a set of RSSI values could be observed from an MMBB, and a set of range values was calculated by means of the BLE units' respective LSPLMs. A range value within three standard deviations ($\mu \pm 3\sigma$) was used to estimate the real distance. Then, some range values were removed by setting up two thresholds. Finally, we considered the average of the remaining values as the estimated range. The estimation algorithm (Algorithm 1) can be summarized as follows:

Algorithm 1. MMBB-Based Range Estimation Algorithm

Input: $\{RSSI_1, RSSI_2, \dots, RSSI_n\}$

Output: P

1. Calculate the ranges via the SLPLMs
2. Select the ranges within three standard deviations ($\mu \pm 3\sigma$).
3. Sort the set of ranges in descending order.
4. Remove top θ percent and bottom ω percent of the range set. The parameters θ and ω are chosen from the range 5%–15%.
5. Average of the remaining ranges P is the estimated range of the MMBB.

Return P

3.2. Pedestrian Walking Speed Measured by Accelerometer

In this paper, we estimated a pedestrian's walking speed using a smartphone built-in accelerometer and used it to constrain the subsequent position estimation results. Firstly, the user's walking events were detected with the help of triaxial accelerometer measurements, and then the step frequency and the user's height were combined to calculate the step length. Finally, the walking speed was estimated from the step frequency and step length:

$$v = \text{stepLength} \cdot \text{stepFrequency} \quad (2)$$

In general, the output of an accelerometer appears as harmonic oscillation waveforms caused by walking behaviors [30], therefore, the cyclic pattern of the accelerometer magnitude can be used in the step detection. Compared with methods such as average cross-counting, cross-correlation, spectral analysis, and thresholding, peak detection methods are computationally less intensive and highly accurate [31].

Consequently, the magnitude of the accelerometer $a_{mag,k}$ was used for peak detection and which can be expressed as:

$$a_{mag,k} = \sqrt{a_{x,k}^2 + a_{y,k}^2 + a_{z,k}^2} \quad (3)$$

where $a_{x,k}$, $a_{y,k}$, and $a_{z,k}$ are the measurements form of the triaxial accelerometer.

In addition, we designed a tenth-order low-pass Butterworth filter with a 3 Hz cut-off frequency to handle the high frequency noise and unstable output of the accelerometer [32].

The two adaptive thresholds were selected from magnitude (vertical axes) and time (horizontal axes) dimensions, as shown in Figure 2 (Right side). The two adaptive thresholds were identified by the following criteria [33]:

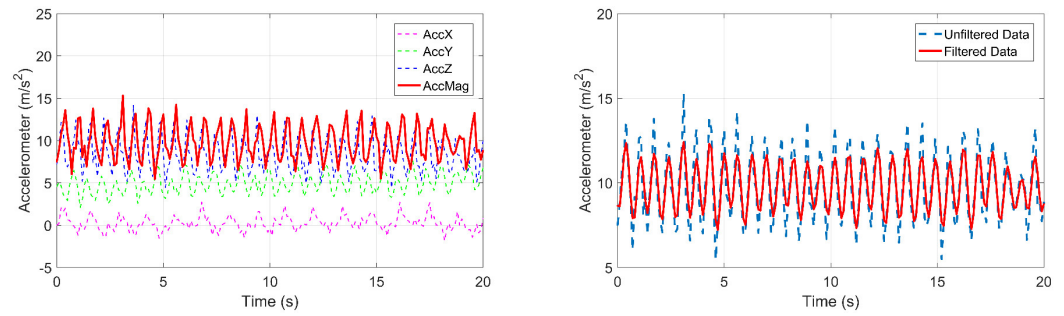


Figure 2. (left side) The cyclic pattern of triaxial and total accelerations during a walking process; (right side) comparison between the total accelerations and total accelerations after filtering.

$$S_k = \begin{cases} \text{step point, } |a_{mag,k}| > \max \left(|a_{mag,win}|, \mu_{a_{mag,k-1}} - \frac{\sigma_{a_{mag,k-1}}}{\alpha} \right) \cap (\Delta t_{min} < T_{S_k} < \Delta t_{max}) \\ \text{intermediate sample, otherwise} \end{cases} \quad (4)$$

where $\mu\gamma$ and $\sigma\gamma$ denote the average and the standard deviation of the magnitude (vertical axis) of the previous steps, respectively, and $\Delta t_{min} \sim \Delta t_{max}$ represents the time interval of human normal walking.

In our previous study [33], we compared commonly used compensation models [34–38] and, in this paper, we chose the empirical linear model proposed in [35] to represent the relationship between pedestrian height, stride frequency, and step length. This model was calibrated by pretraining, using real-time walking distances [27].

3.3. Tightly Coupled Integration of MMBB Range and Accelerometer Integration System

Tightly coupled (TC) architecture was implemented to integrate the MMBB and accelerometer and an overview of the proposed integrated method is given in Figure 3. As shown in Figure 3, the integration system has two measure channels. One channel is the resolving channel of the MMBB, in which the LSPLM converts the measured RSSI values of a smartphone to the range between the MMBB and a pedestrian. The other channel is the pedestrian walking speed estimating channel, in which the accelerometer measured data is converted to the walking speed of a pedestrian. Finally, combining two types of observations, an unscented Kalman filter (UKF) method estimates the system state, including the location and speed.

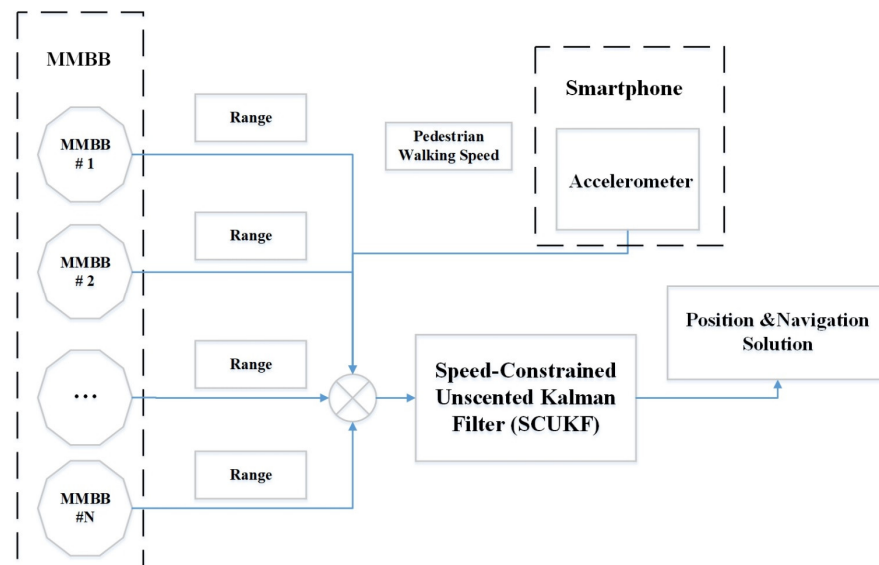


Figure 3. Tightly coupled integration.

3.3.1. Dynamic Model

In the proposed speed constrained UKF integration method, 3D position and velocity for tracking purposes are selected as the system state, $[E_i \ N_i \ U_i]$ is the position vector in the East, North and Up (ENU) reference frame; $[v_{E,i} \ v_{N,i} \ v_{U,i}]$ is the velocity vector in the ENU reference frame; we defined the state vector as $\hat{x}_i = [E_i \ N_i \ U_i \ v_{E,i} \ v_{N,i} \ v_{U,i}]^T$. The system state transfer function f_h is as follows:

$$\hat{x}_{h,i} = f_h(\hat{x}_{h,i-1}, u_i) + w_{h,i-1} \quad (5)$$

$$f_h(\hat{x}_i, u_k) = \begin{bmatrix} E_i + \Delta t \cdot v_{E,i} \\ N_i + \Delta t \cdot v_{N,i} \\ U_i + \Delta t \cdot v_{U,i} \\ v_{E,i} \\ v_{N,i} \\ v_{U,i} \end{bmatrix} \quad (6)$$

where $f_h(\cdot)$ represents the system dynamic model, system input u_i . The process noise $w_{h,i}$ drives the dynamic system, Δt denotes the sample intervals.

3.3.2. Observation Model

As we have two different types of sensor technologies available, the observation model depends on the specific technology used.

Range Observation Model: Range is not a direct observation, but rather a conversion from signal strength to range by using a path loss model. In Section 2, we proposed the range estimation algorithms based on SLPLM and MMBB in detail, and therefore they are not described again. The range observation vector is defined as follows:

$$z_i^{dis} = [pr_{1,i} \ pr_{2,i} \ \cdots \ pr_{m,i} \ \cdots \ pr_{M,i}]^T \quad (7)$$

The range observation model is nonlinear and defined as:

$$z_{r,i} = h_{r,h}(\hat{x}_i, u_i) + v_{h,i}^r \quad (8)$$

$$h_{r,h}(\hat{x}_i, u_k) = \begin{bmatrix} Dis(\hat{x}_i, p_1) \\ Dis(\hat{x}_i, p_2) \\ \vdots \\ Dis(\hat{x}_i, p_m) \\ \vdots \\ Dis(\hat{x}_i, p_M) \end{bmatrix} \quad (9)$$

$$Dis(x_{h,k}, p_m) = \sqrt{(E_{p,m} - E_k)^2 + (N_{p,m} - N_k)^2 + (U_{p,m} - U_k)^2} \quad (10)$$

where $h_{r,h}(\cdot)$ represents the range of observation function, $p_m = [E_{p,m}, N_{p,m}, U_{p,m}]$ denotes the MMBB location in the ENU reference frame, and u_i represents system input.

Walking Speed Observation Model: By using a smartphone with a built-in accelerometer, the pedestrian's walking speed is estimated and used to constrain the state of the system. The speed observation vector is nonlinear and defined as:

$$z_i^v = [v_i]^T \quad (11)$$

The speed observation model is nonlinear and defined as:

$$z_{v,i} = h_{v,h}(\hat{x}_i, u_i) + v_{h,i}^v \quad (12)$$

$$h_{v,h}(\hat{x}_i, u_i) = \left[\sqrt{v_{E,i}^2 + v_{N,i}^2 + v_{U,i}^2} \right] \quad (13)$$

where $h_{r,h}(x_i, u_i)$ represents the walking speed observation function and u_i denotes system input.

The Kalman framework allows for the fusion of various types of sensors, the observation vector is specified:

$$z_i = \begin{bmatrix} z_i^{dis} & z_i^v \end{bmatrix}^T \quad (14)$$

It is noteworthy that the dimension of this observation vector z_i^{dis} may vary accordingly with the number of observed MMBBs. In addition, the observation of pedestrian walking speed can be achieved based on the accelerometer, which can use high-frequency sensors to detect the motion as long as a smartphone is turned on.

3.3.3. Pedestrian Walking Speed Constrained Unscented Kalman Filter

In the above derivation, we discuss a discrete-time system represented by:

$$\begin{cases} x_{h,i+1} = f_h(\hat{x}_{h,i}, u_k) + w_{h,i} \\ z_{r,i} = h_{r,h}(\hat{x}_i, u_k) + v_{h,i}^r \end{cases} \quad (15)$$

where $f_h(\cdot)$ represents the system dynamic model; $h_{r,h}(\cdot)$ is the nonlinear function describing the observation model; system input u_i , the process noise $w_{h,i}$, and $v_{h,i}^r$ are the uncorrelated zero-mean Gaussian white noise processes with covariances.

The procedure of the speed constrained unscented Kalman filter (SCUKF) can be described as follows:

Step 1 Initialization

The state estimate $\hat{x}_{i-1}$ and the error covariance matrix P_{i-1}^+ is given.

Step 2 Generate sigma points

A total of $2L + 1$ sigma points are calculated from the columns of matrix X_{i-1}^a as:

$$X_{i-1}^a = \begin{bmatrix} \hat{x}_{i-1}^+ & \hat{x}_{i-1}^+ \pm \sqrt{(L + \lambda)P_{i-1}^+} \end{bmatrix} \quad (16)$$

$$\begin{cases} X_{i-1}^0 = \hat{x}_{i-1}^+ \\ X_{i-1}^{(a)} = \hat{x}_{i-1}^+ + \sqrt{(L + \lambda)P_{i-1}^+} & a = 1, \dots, L \\ X_{i-1}^{(a)} = \hat{x}_{i-1}^+ - \sqrt{(L + \lambda)P_{i-1}^+} & a = L + 1, \dots, 2L \end{cases} \quad (17)$$

with weights as:

$$\begin{cases} w_0^m = \lambda / (L + \lambda) \\ w_0^c = \lambda / (L + \lambda) + (1 - \alpha^2 + \beta) \\ w_j^m = w_j^c = 1 / \{2(L + \lambda)\} & j = L + 1, \dots, 2L \end{cases} \quad (18)$$

Step 3 Time update

The sigma points yielded through the Equation (5) are as follows:

$$X_{i|i-1}^x = f(X_{i-1}^x, X_{i-1}^v) \quad (19)$$

The transformed samples $X_{i|i-1}^x$ are used to capture the specific information on the distribution of predicted state. The the predicted state mean and covariance of the transformed samples are estimated as follows:

$$\hat{x}_i^- = \sum_{j=0}^{2L} w_j^m X_{j,i|i-1}^x \quad (20)$$

$$P_i^- = \sum_{j=0}^{2L} w_j^c (X_{j,i|i-1}^x - \hat{x}_i^-)(X_{j,i|i-1}^x - \hat{x}_i^-)^T \quad (21)$$

where L is the dimension of the augmented state.

The transformed sigma points yielded by the Equation (12) are as follows:

$$Z_{i|i-1} = h(X_{i|i-1}^x, X_{i-1}^n) \quad (22)$$

The predicted measurement is computed as follows:

$$\hat{z}_i^- = \sum_{j=0}^{2L} w_j^m Z_{j,i|i-1} \quad (23)$$

Step 4 Measurement update

The predicted measurement covariance is computed as follows:

$$P_i^h = \sum_{j=0}^{2L} w_j^c (Z_{j,i|i-1} - \hat{z}_i^-)(Z_{j,i|i-1} - \hat{z}_i^-)^T \quad (24)$$

The cross-correlation covariance matrix between the predicted state and measurement is given by:

$$P_i^{fh} = \sum_{j=0}^{2L} w_j^c (X_{j,i|i-1}^x - \hat{x}_i^-)(Z_{j,i|i-1} - \hat{z}_i^-)^T \quad (25)$$

The Kalman gain is calculated by:

$$\kappa_i = P_i^{fh} (P_i^h)^{-1} \quad (26)$$

Then the state estimate $\hat{x}_i^+$ and the corresponding error covariance matrix P_i^+ can be updated by:

$$\hat{x}_i^+ = \hat{x}_i^- + \kappa_i (z_i - \hat{z}_i^-) \quad (27)$$

$$P_i^+ = P_i^- + \kappa_i P_i^h \kappa_i^T \quad (28)$$

Step 5 Time update: return to Step2

In summary, the SCUKF can be summarized as follows (Algorithm 2):

Algorithm 2.

Pedestrian Walking Speed Constrained Unscented Kalman Filter Algorithm

Input: $\{RSSI_1, RSSI_2, \dots, RSSI_n\}$

Output: P

Set $\hat{x}_{i-1}$ and P_{i-1}^+ .

For $i=1$ to N

Generate sigma points:

 Generate the sigma points according to Equation **Error! Reference source not found.**, **Error! Reference source not found.** and **Error! Reference source not found.**.

Time update:

- 1: **Evaluate** the sigma points with the dynamic model function according to Equation **Error! Reference source not found.**.
- 2: **Evaluate** the predicted state mean $\hat{x}_i^-$ and error covariance P_i^- according to Equation **Error! Reference source not found.** and **Error! Reference source not found.**.
- 3: **Evaluate** the generate d sigma points with measurement function according to Equation **Error! Reference source not found.**.
- 4: **Estimate** the predicted measurement according to Equation **Error! Reference source not found.**.

Measurement update

-
- 1: **Estimate** the innovation covariance matrix P_i^h according to Equation
Error! Reference source not found..
 - 2: **Estimate** the cross-covariance matrix P_i^{fh} according to Equation
Error! Reference source not found..
 - 3: **Calculate** the Kalman gain K_i according to Equation
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 - 4: **Estimate** the updated state according to Equation **Error! Reference source not found..**
 - 5: **Estimate** the updated error covariance according to Equation
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- End**
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4. Experimental Assessment

4.1. Setup

The performance of the proposed indoor positioning algorithm was evaluated based on field tests carried out in the Lounge Café at the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS) at Wuhan University. As Figure 4 shows, the test area is 123.18 square meters ($7.67 \text{ m} \times 16.06 \text{ m}$). Six MMBBs were deployed in the lounge. Four smartphones, including Huawei Mate9, P9, P9 plus, and Honor 8 were used for the experiment, all with Android platforms. The sampling rate of the accelerometers in the smartphones was 100 Hz. The fireld test were done on the Android application in real time. A reference point, used as the ground truth, was measured using a laser distance meter measurer with 1.5 mm precision.

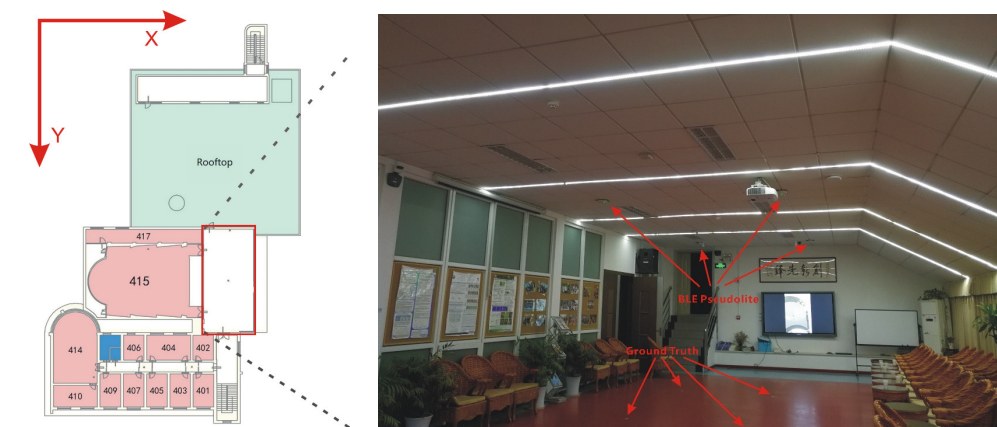


Figure 4. The experimental site.

The experiment consisted of two parts. In the first part, we compared the performance of the proposed finite range estimation method based on MMBB and SLM to a state-of-art model, including static and dynamic tests. In the second part, we evaluated the proposed SCUKF indoor positioning algorithm. We also compared the classic triangulation and UKF methods with the SCUKF method in the work.

To assess performance, we compared the proposed LSPLM with the model given by radius networks [39]. The tests were carried out on the playground of Wuhan University, where there is open ground and the multipath effect is the lowest. During the testing, 72 BLE units were used and the RSSI values were collected by the four smartphones.

According to [39], the three constants in radius networks are 0.89976, 7.7095, and 0.111. The constants of the proposed LSPLM were calibrated by using the calibration method. The parameters θ and ω in the range estimation algorithm were set to be 30 and 25, respectively.

The second part of the experiment was to evaluate the accuracy of the MMBBs-based ranging algorithm in an indoor environment. Here, we compared the accuracy of MMBB-based ranging and BLE unit-based ranging with different PLMs. The walking speed of the participant fell to 1.0–2.0 m/s.

The accuracy of the proposed SCUKF indoor positioning algorithm base on MMBB was investigated and compared with the conventional UKF algorithm and the LST method. For the step detection method, values of 0.25, 2, and 0.334 were assigned to α , $\Delta t_{\max}$, and $\Delta t_{\min}$, respectively. The coefficients $a = 0.371$, $b = 0.227$, and $c = 1$ were the three parameters of the step length model. For the conventional UKF method, a 3D position was chosen as the system state, and the observation vector consisted of estimated ranges. The initial covariance P and additional positive definite matrix ΔQ for position estimation were set to $32 \times I^2$ and $0.12 \times I^2$, respectively. In this part, two tests were carried out in the Lounge Café with different walking speeds. Test 1 lasted for about 4 min, while Test 2 lasted for 6 min at a relatively faster speed.

4.2. Experimental Results and Analysis

First, we compared the performance of the ranging methods under static tests outdoors. In Figure 5, the resulting cumulative distribution functions (CDFs) for the same RSSI value dataset are given, the mean ranging error of the radius networks PLM is 3.21 m, and the 50% and 95% error is 1.55 m and 12.00 m, respectively. These are much larger than the errors of the proposal LSPLM, which are 0.94 m, 5.01 m, and 0.50 m. The results of standard deviations and variance of the ranging error are also listed in Table 1.

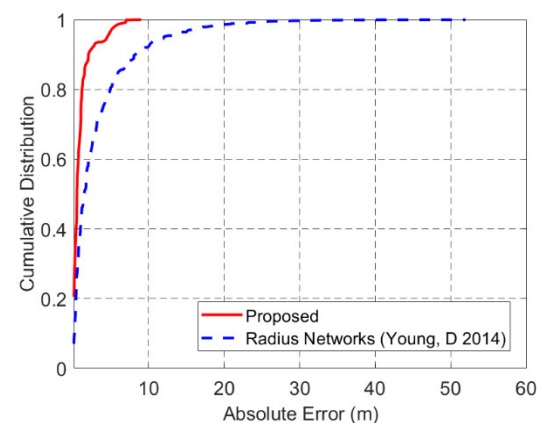


Figure 5. Cumulative error percentages in different cases.

Table 1. The comparison of static ranging errors.

Stat.	Radius Networks (Young, D 2014)	Proposed
Mean (m)	3.21	0.94
Std (m)	4.59	1.35
Var (m ²)	21.12	1.82
95th (m)	12.00	5.01
Median (m)	1.55	0.50

The above ranging results demonstrate that the proposed LSPLM can effectively improve the performance of the RSSI-based ranging. Therefore, it is clear that the proposed LSPLM has achieved better accurate and stable results. The reason is that the LDPLM is

more susceptible to signal transition and multipath effects than the linear model. Especially, at the position far from the BLE unit, the RSSI values are small, and a small distortion can lead to large errors.

Then, we compared the ranging performance based on multiple modules and single modules under indoor dynamic conditions. Figure 6 shows the ranging errors obtained by three methods, including the single BLE module-based method and the MMBB-based method using different PLMs. During this period, the ranging errors of the single module-based with radius networks model are within 9.7240 m and 57.8789 m; the 50% and 95% ranging errors are at 5.9170 m and 33.8746 m, respectively. The ranging errors are much larger than that of the single module-based with the LSPLM method, which are within 0.0097 m and 7.8047 m; the 50% and 95% ranging errors are at 2.7362 m and 5.7604 m, respectively. It is clear that the MMBB-based method provides the best ranging result. The ranging errors obtained by the method are within 1.4358 m and 5.1258 m; the 50% and 95% ranging errors are at 1.2216 m and 3.5351 m, respectively. The result of the means, standard deviations, and variance of the ranging error are also listed in Table 2.

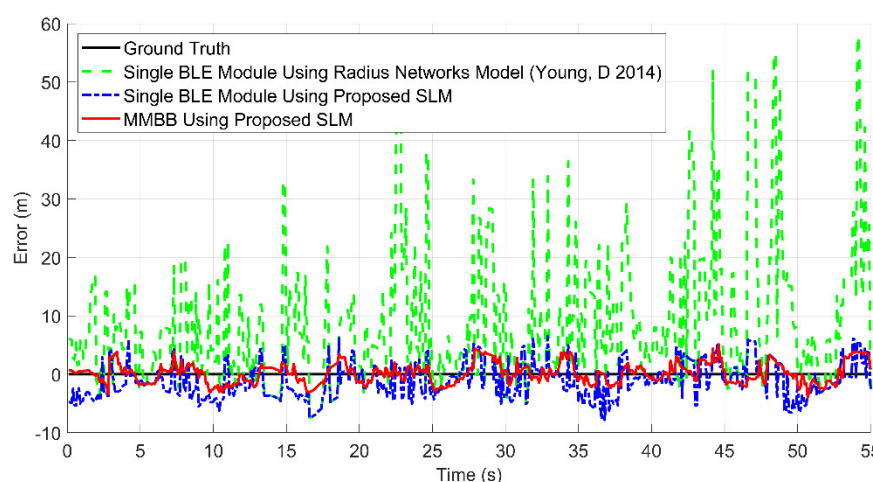


Figure 6. Comparison of ranging error versus time epoch between BLE unit-based and MMBB-based methods.

Table 2. The comparison of dynamic ranging errors.

Stat.	Single BLE Module Using (Young, D 2014)	Single BLE Module Using SLM	Proposed
Mean (m)	9.72	2.79	1.43
Std (m)	11.15	1.74	1.03
Var (m ²)	124.34	3.04	1.07
Max (m)	57.87	7.80	5.12
95th (m)	33.87	5.76	3.53
Median (m)	5.91	2.73	1.22

On the basis of the results, the accuracy of the MMBB-based ranging method is much higher than the common ranging method, and it is obviously better than the unit-based method. In the indoor environment, the ranging method has good robustness, which is due to the following three reasons: First of all, the BLE module array in circle arrangement provides several observations in one positioning epoch. Second, from the first experimental results, the proposed LSPLM can significantly reduce the impact of signal transition and multipath effects. Third, the proposed algorithm based on statistics can further weaken the impact of signal transition and multipath effects by using two empirical thresholds.

Figure 7 shows the positioning results from Test 1 and Test 2. In detail, Figure 7a,b show the absolute error versus positioning epoch of three different algorithms. Figure 7c,d demonstrate the cumulative distribution functions (CDFs) of the three algorithms. Based on the MMBB ranging algorithm, the mean error of the three algorithms (LST, UKF, and proposed SCUKF) is less than 2 m. In Test 1, the position error of the LST are within 0.1627 m and 3.9059 m; the 50% and 95% position errors are at 1.4858 m and 2.9903 m. These are larger than the position errors of the UKF method, which are within 0.0607 m and 2.7712 m; the 50% and 95% position errors are at 0.8886 m and 2.1606 m, respectively. The best accuracy obtained by the proposed SCUKF is within 0.0840 m and 2.0072 m; the 50% and 95% position errors are at 0.7989 m and 1.3978 m, respectively. In Test 2, the position errors of the LST are within 0.1380 m and 6.2886 m; the 50% and 95% position errors are at 1.7846 m and 1.9389 m, respectively. These are larger than the position errors of the UKF method, which are within 0.1222 m and 4.2963 m; the 50% and 95% position errors are at 0.8958 m and 0.9635 m, respectively. Therefore, it is clear that the best accuracy obtained by the proposed SCUKF is within 0.0793 m and 2.6822 m; the 50% and 95% position errors are at 0.7550 m and 0.8143 m. The means, standard deviations, and variance of position errors are listed and compared in Table 3.

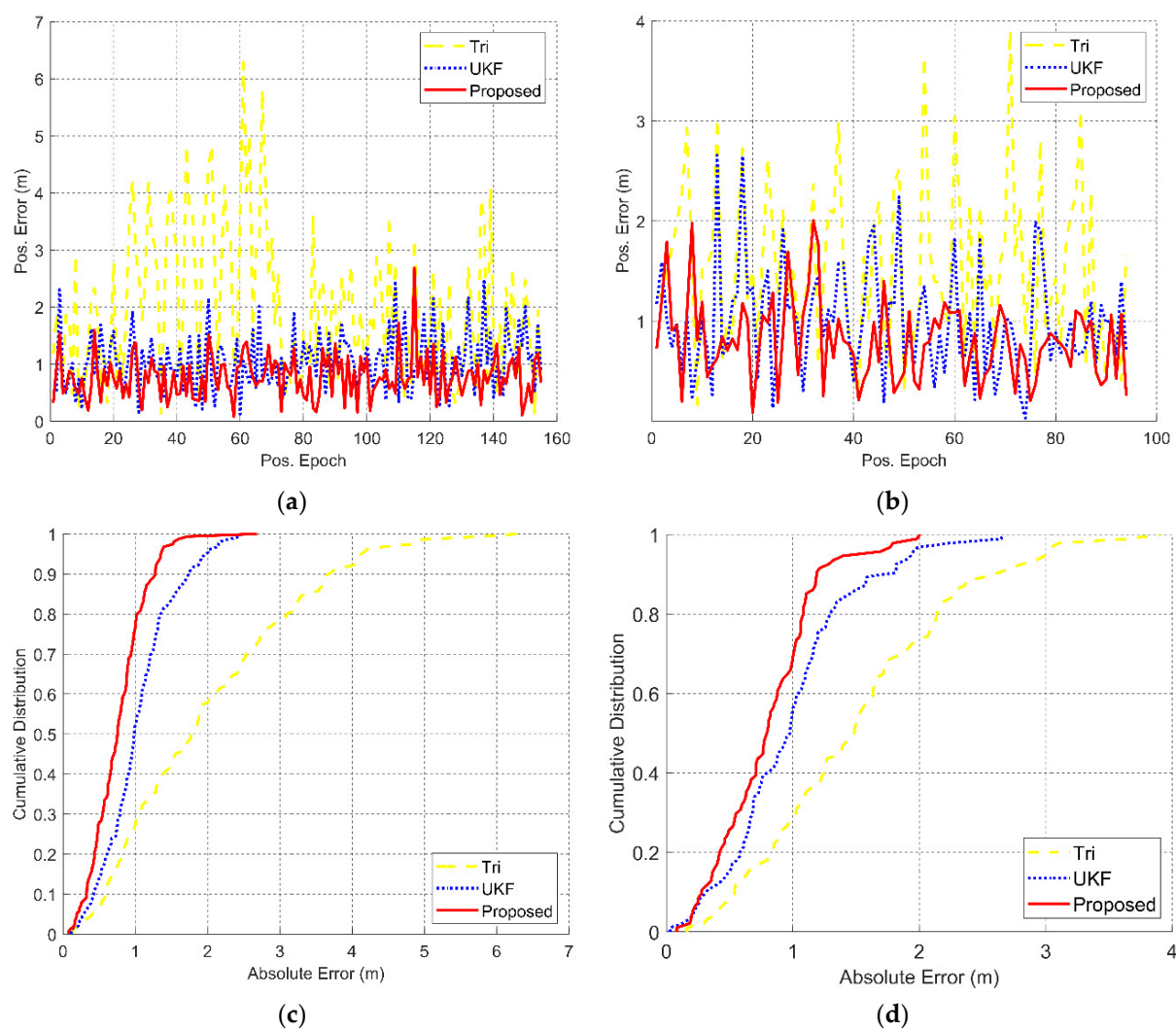


Figure 7. Comparison of ranging error versus time epoch between BLE unit-based and MMBB-based methods. (a) Positioning error of Test 1 (b) Positioning error of Test 2 (c) Cumulative error percentages of Test 1 (d) Cumulative error percentages of Test 2.

Table 3. Position error comparison.

Test	Stat.	Triangulation	UKF	Proposed
1	Mean (m)	1.99	1.02	0.77
	Std (m)	1.25	0.63	0.37
	Var (m ²)	1.58	0.41	0.14
	Max (m)	6.28	4.29	2.68
	95th (m)	1.93	0.96	0.81
	Median (m)	1.78	0.89	0.75
	Min (m)	0.13	0.12	0.07
2	Mean (m)	1.52	0.95	0.80
	Std (m)	0.78	0.52	0.39
	Var (m ²)	0.62	0.27	0.15
	Max (m)	3.91	2.77	2.01
	95th (m)	2.99	2.16	1.39
	Median (m)	1.48	0.88	0.79
	Min (m)	0.16	0.06	0.08

From the results based on the MMBB, the mean error of the three algorithms (LST, UKF, and proposed SCUKE) is less than 2 m. The mean and standard deviation error of the proposed SCUKE is lower than those of LST and UKF, and it obtained the best accuracy and stability. One reason is that, as compared with the UKF algorithm, the LST only minimizes the square of the measurement error, while the UKF algorithm takes into account the dynamic measurement and the dynamic changes in the system. The system state was predicted by using the dynamic models and combined with measurement based on weight. In this way, the UKF not only minimizes the square of the measurement error, but also minimizes the estimated error variance matrix. The other reason is that the observation model of SCUKE is one more pedestrian speed than that of UKF. This additional observation constrains the system state from the speed dimension. In other words, the positioning results are constrained by the speed measurement within a reasonable range of pedestrian walking speed.

5. Conclusions

In this paper, we introduced an MMBB-based indoor positioning solution. It realized a tightly coupled fusion architecture implemented on a smartphone, which integrated the MMBBs and accelerometer. It is an instant positioning solution without any training phase by adopting a calibrated linearly segmented model for ranging. In addition, the application of pedestrian walking speed derived by a smartphone accelerometer is used to constrain an unscented Kalman filter method for estimating the location and speed. Field tests were carried out to verify the proposed SLM, ranging and positioning algorithms. The proposed SLM was evaluated and compared with a state-of-art PLM by field tests in an open environment. The performance of the SLM is about 1 m in terms of the mean error. The error obtained by the MMBB-based ranging algorithms is 1.43 m in the real indoor environment with normal walking speed. The performance of the positioning algorithm was effective and achieved the best accuracy as compared with the least square triangulation and unscented Kalman filter estimation methods. Finally, the accuracy of the proposed speed constrained unscented Kalman filter method is 0.79 m in terms of horizontal error distance; the positioning frequency and the time to first fix are 10 Hz and 100 milliseconds, respectively. The coordinate frame of the system is compatible with GNSS signals, which makes the system work seamlessly with the GNSS. In future work, we plan to focus on improving the scope and accuracy of range estimation based on MMBB hardware and the robustness of the localization system. In addition, we plan to fuse the walking speed of pedestrians with other measurement data for indoor localization, such as geomagnetic, Wi-Fi RTT range, and acoustic signal.

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